Table of Contents

Copyright

Brief Table of Contents

Table of Contents

Preface

Acknowledgments

About This Book

About the Authors

About the Cover Illustration

1. Foundations

Chapter 1. What is reinforcement learning?

1.1. The “deep” in deep reinforcement learning

1.2. Reinforcement learning

1.3. Dynamic programming versus Monte Carlo

1.4. The reinforcement learning framework

1.5. What can I do with reinforcement learning?

1.6. Why deep reinforcement learning?

1.7. Our didactic tool: String diagrams

1.8. What’s next?

Summary

Chapter 2. Modeling reinforcement learning problems: Markov decision processes

2.1. String diagrams and our teaching methods

2.2. Solving the multi-arm bandit

2.2.1. Exploration and exploitation